

Ernest Choi

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Projects

Algorithm Terrain Visualizer | *JavaScript, C++, Java* ernestchoi.com/terrain

- Built a 3D terrain generator in **JavaScript** on **WebGLRenderer**, applying procedural texture algorithms (Perlin, Gaussian) prototyped in **C++** and **Java** to visualize complex math concepts.
- Developed a voxel-style tile system with adjustable generation parameters, dynamic lighting and shadows, useful for explaining physics and geometry to students in a teaching role.

AI Resume & Job-Search Automation Bot | *Python* github.com/ErChoi-com/DiscordBot

- Built a **Discord.py** bot in **Python** that scrapes job postings through **REST APIs**.
- Designed an **LLM**-driven resume pipeline with fallback across multiple providers, prompting for structured **JSON** output, a skill transferable to providing clear instructions to students in a learning context.

Experience

Engineering Instructor, Obotz Robotics – Markham, Ontario September 2022 – Present

- Taught digital logic concepts and embedded **C** programming to High School students, raising robotics problem-completion scores by 35% through focused learning and practice.
- Coached students in robotics competitions to place in the 75th percentile, helping them apply science and math principles with custom **Arduino** hardware.
- Communicated students' technical progress to parents in clear, non-technical terms, fostering a supportive learning environment at the **Centre** for teaching.

Technical Development Intern, City of Markham – Markham, Ontario July 2022 – August 2024

- Maintained clear **LaTeX** documentation of website workflows and operating processes, helping cut website downtime by 41%.
- Tracked project milestones and reported progress to management using **Microsoft Suite**, reducing coordination delays by 14%, which can aid in helping students manage their time and learning.
- Validated and tested system changes against quality standards, reducing reported errors by 20%.

Frontend Development Intern, MCG3D – Remote January 2026 – March 2026

- Built production web applications in **React** and **TypeScript**, focusing on a scalable component architecture and fast, user-centric interfaces for effective learning.
- Integrated frontend features with **Python Flask** backend services over secure **RESTful APIs**, including AI-enabled workflows, a systematic approach for helping users.

Machine Learning Intern, Goppter Commerce Solutions – Remote September 2025 – December 2025

- Engineered a distributed, low-latency data pipeline for financial modeling, streaming **Kafka** data into asynchronous **Python** services using **Pandas** and **NumPy**.

Skills

Languages: JavaScript/Node.js, Java, Python, C++, C, SQL, HTML, CSS

Frameworks/Libraries: OpenGL, WebRTC, JavaFX, React, TypeScript, Flask, FastAPI, TensorFlow

Tools: Git, PostgreSQL, Ubuntu Linux, Docker, Kafka, Tableau, Power BI, FPGA

Platforms: GitHub, GitLab, Raspberry Pi, Arduino

Relevant Courses: Object Oriented Eng Analysis and Design, Digital Systems, Electronic Circuits I, Microprocessor Systems

Education

Toronto Metropolitan University – Bachelor of Engineering in Computer Engineering 2023–2027